SPECIFICATIONS OF SFC0032_16B9_HE[1]

Rev 0.1

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[1] This specification may be revised without notice.

TSMC **SFC0032_16B9_HE**

32K X 16 **Embedded Flash Memory**

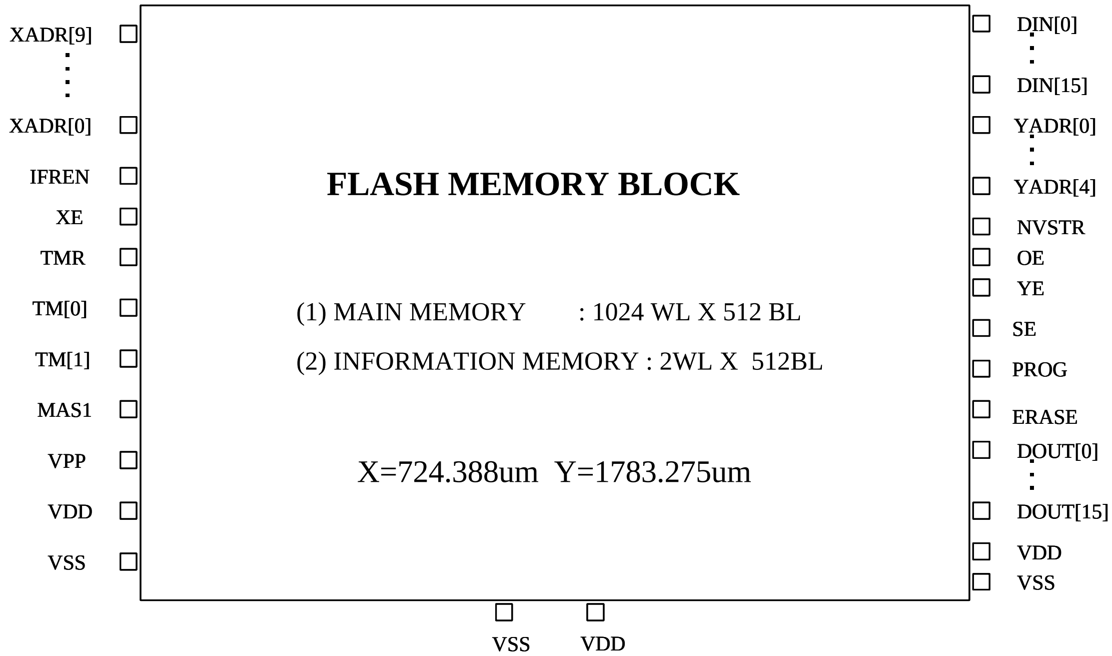
General Description

The SFC0032_16B9_HE is a CMOS page erase, word(16-bit) program Embedded Flash Memory which is partitioned into two memory blocks. One is the main memory block, the other is the information block .The main memory block is organized as a 32768-word by 16-bit. The information block is organized as a 64-word by 16-bit . The information block can be for the storage of device information. The page erase operation erases all bytes within a page. A page is composed of eight adjacent rows for main memory block and two adjacent row for information block .The split gate cell design and thick oxide tunneling injector attain better reliability and manufacturability compared with alternative approaches. The SFC0032_16B9_HE erases and programs with a 2.5-volt only power supply.

Features

- Single 2.5-volt power supply
- Memory Organization 32K×16 + 64×16
- Page Erase Capability: 512 bytes per page
- Access Time : 20ns(max)
- Current
  Operating : 12mA(max)
  Standby : 10uA(max)
- Endurance: 20,000 Cycles(min)
- Greater than 100 years Data Retention under room temperature.
- Fast Page Erase/Word Program Operation
- Word Program Time: 20us(min)
  Page Erase Time: 20 ms(min)
  Mass Erase Time: 200 ms(min)

Pin Assignments



Connector Description

User Mode[1]

Connector name	Direction	Description
XADR[9:0]	I	X address input, access rows, XADR[2:0] select one row within a page of main memory block, XADR[0] select one row within a page of information block.
YADR[4:0]	I	Y address input, access a byte within a row.
DIN[15:0]	I	Data input bus.
DOUT[15:0]	O	Data output bus.
XE	I	X address enable, all rows are disabled when XE=0.
YE	I	Y address enable, YMUX is disabled when YE=0.
SE	I	Sense amplifier enable.
IFREN[3]	I	Information block enable
OE	I	Output enable, tri-state DOUT when OE=0.
ERASE	I	Defines erase cycle.
MAS1	I	Defines mass erase cycle, erase whole block.
PROG	I	Defines program cycle.
NVSTR	I	Defines non-volatile store cycle.
VDD[2]	I	Power supply
VSS[2]	I	Ground

Notes:

1. Need to multiplex all control, address and data signals to package pins to facilitate direct memory test.
2. If routing power width is as wide as 40um, connecting only one set of VDD/VSS is allowable. If not, the total width of power line connecting to two sets or three sets of VDD/VSS should not be less than 40um.
3. In order to access the information block, the XADR[2:1] must be kept low, the XADR[9:3] are don't-care. XADR[0] is used to select one row within the information block.

Test Mode[1]

Connector name	Direction	Description
TMR[2]	I	Register reset for test mode. In user mode, it should be connected to VDD.
VPP	I/O	High voltage pin used in test mode and operating in 0~15V. A high voltage pad with a I/O path for VPP pin and a input path for input pin will be provided by TSMC. So this high voltage pad can be used as a normal input pad in user mode. But the input path has a Vt drop when pass logic high(VDD) signal.
TM[1:0]	I/O	Analog pins used in test mode and operating in 0~VDD. These analog pins are high impedance in user mode.

Notes:

1. Need to multiplex the test mode related signals to package pins for memory testing.
2. TMR is a digital pin and operating in 0~VDD.

Recommended DC Operating Conditions

(T_J = 0 ºC to 125 ºC)

Parameter	Symbol	Min.	Typ.	Max.	Unit
Supply Voltage	V_{DD}	2.25	2.50	2.75	V
Ground	V_{SS}	0	0	0	V

DC Electrical Characteristics (T_J = 0 ºC to 125 ºC, V_{DD} = 2.25 V to 2.75 V, V_{SS} = 0V)

Parameter	Symbol	SFC0032_16B9_HE		Unit	Condition
		Min.	Max.		
Operating Current Read Program Erase Mass erase	I_{DD_1}	 - - - -	 12 7 5 5	mA	Min.Cycle,Duty= 100% V_{IN} = V_{DD} or V_{SS} $I_{I/O}$ = 0mA
Static Read Current	I_{DD_2}	-	7	mA	XE=YE=SE=OE=VDD $I_{I/O}$ = 0mA
Standby Power Supply Current	I_{SB}	-	10	uA	XE, YE, SE, OE, PROG, ERASE, and MAS1 are all V_{SS}. Other V_{IN} = V_{DD} or V_{SS}

Truth Table

Mode	XE	YE	SE	OE	PROG	ERASE	MAS1	NVSTR
Standby	L	L	L	L	L	L	L	L
Read	H	H	H	H	L	L	L	L
Word Program	H	H	L	L	H	L	L	H
Page Erase	H	L	L	L	L	H	L	H
Mass Erase	H	L	L	L	L	H	H	H

IFREN Truth Table

Mode	IFREN=1	IFREN=0
Read	Read information block	Read main memory block
Word Program	Program information block	program main memory block
Page erase	Erase information block	Erase main memory block
Mass erase	Erase both block	Erase main memory block

Test Mode Decoder Truth Table

Mode	CODE				
	ERASE	YE	XE	IFREN	MAS1
Test mode 1	L	H	L	L	H
Test mode 2	L	H	L	H	L
Test mode 3	H	H	H	L	L
Test mode 4	H	H	H	L	H
Test mode 5	H	H	H	H	L

Capacitance

Parameter	Symbol	Min.	Max.	Unit
Address & Control Input pin Capacitance	C_{IN}	-	0.3	pF
Data Input pin Capacitance	C_{DIN}	-	0.3	pF
Data Output pin Capacitance	C_{DOUT}	-	0.1	pF

Timing Parameters [1] (T_J = 0 ºC to 125 ºC, V_{DD} = 2.25 V to 2.75V, V_{SS} = 0V, C_{LOAD}=1pF)

User mode

Parameter	Symbol	SFC0032_16B9_HE		Unit
		Min.	Max.	
X address access time	T_{xa}	-	20	ns
Y address access time	T_{ya}	-	20	ns
OE access time	T_{oa}	-	5	ns
PROG/ERASE to NVSTR set up time	T_{nvs}	5	-	us
NVSTR hold time	T_{nvh}	5	-	us
NVSTR hold time(mass erase)	T_{nvh1}	100	-	us
NVSTR to program set up time	T_{pgs}	10	-	us
program hold time	T_{pgh}	20	-	ns
program time	T_{prog}	20	40	us
address/data set up time	T_{ads}	20	-	ns
address/data hold time	T_{adh}	20	-	ns
recovery time	T_{rcv}	1	-	us
cumulative program HV period[2]	T_{hv}	-	4	ms
Erase time	T_{erase}	20	-	ms
Mass erase time	T_{me}	200	-	ms

Notes:

1. All electrical and timing information listed above are based on SPICE(or equavelent) simulations and subject to change after silicon verification.
2. T_{hv} is the cumulative high voltage programming time to the same row before next erase and the same address can not be programmed more than twice before next erase.
3. All signals that align together in the timing diagrams should be derived from the same clock edge. Set up and hold times are not critical if they are within 10ns.
4. Reading operation is combinatorial. Sequence of timing edges are not important. The latest valid signal determines the access time.
5. All timing measurements are from the 50% of the input to 50% of the output.
6. All input waveforms have rising time(tr) and falling time(tf) of 1ns.
7. For capacitive loads greater than 1pF, access time will increase by 1ns per pF of additional loading.

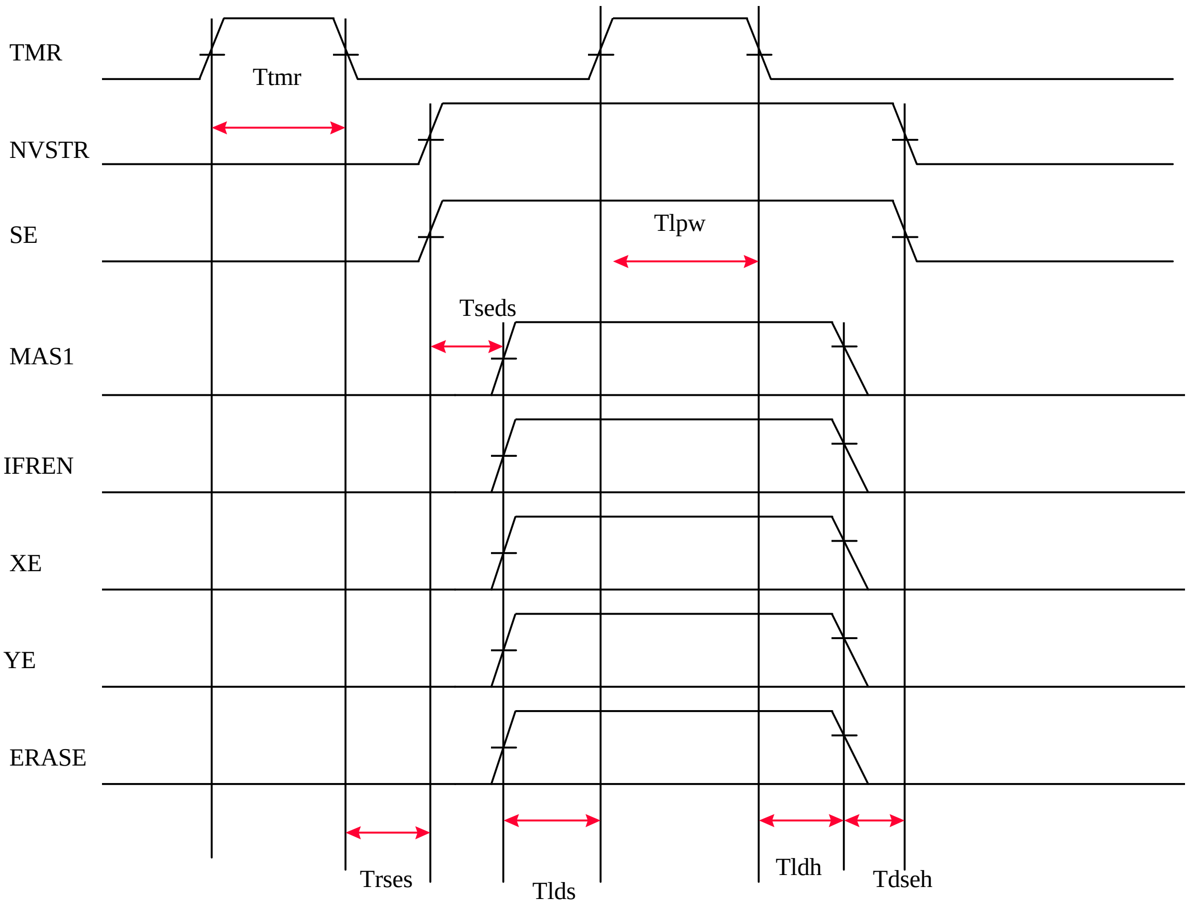
Test mode

Parameter	Symbol	Value		Unit
		Min	**Max**	
TMR pulse width[1]	Ttmr	20	-	ns
TMR to internal TMSE set up time	Trses	10	-	ns
Internal TMSE to data set up time	Tseds	10	-	ns
Data set up time for latch	Tlds	10	-	ns
Set latch pulse width	Tlpw	20	-	ns
Data hold time for latch	Tldh	10	-	ns
Data to internal TMSE hold time	Tdseh	10	-	ns

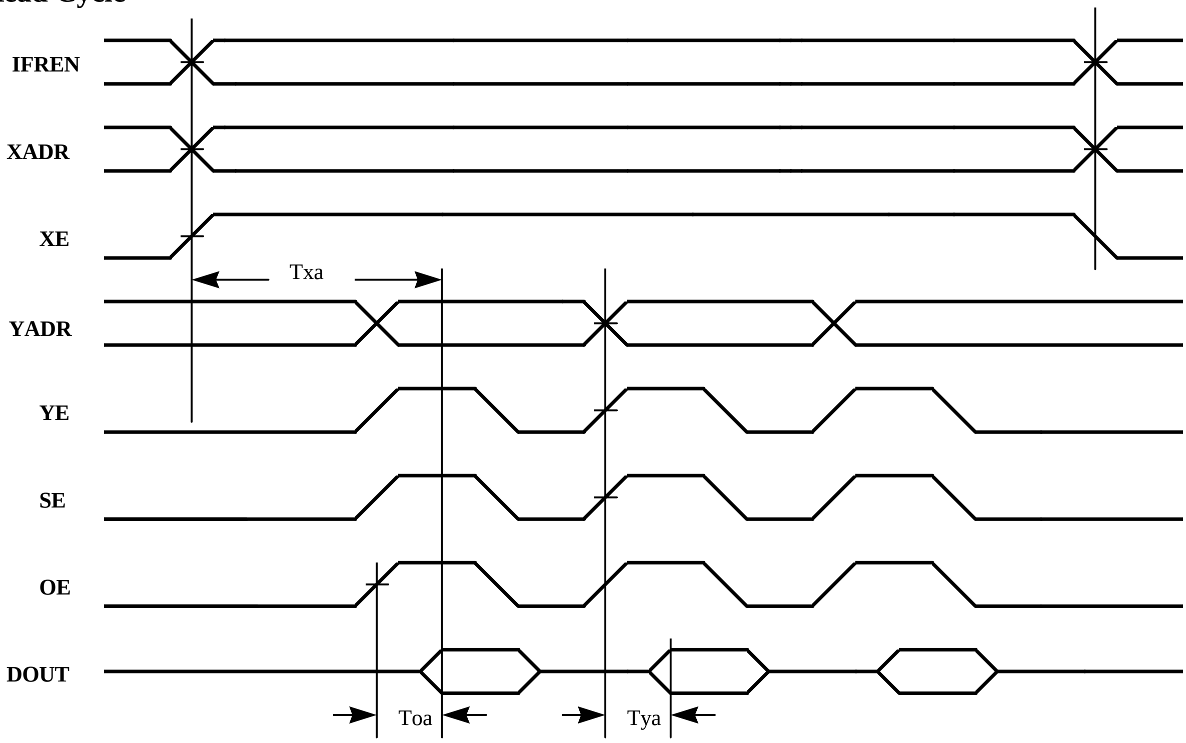
Notes:
0.TMR pulse is for resetting the test mode latch only and it is not required for entering test mode.

Timing Waveforms

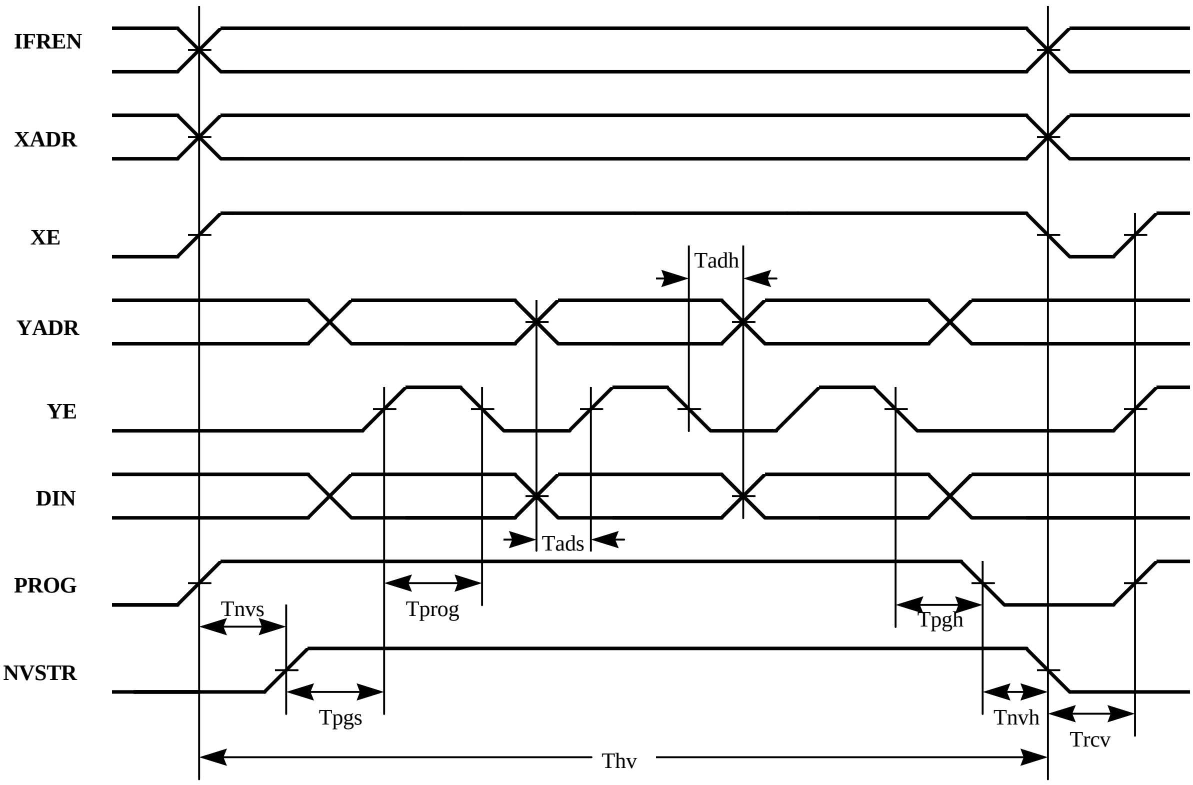
Test mode



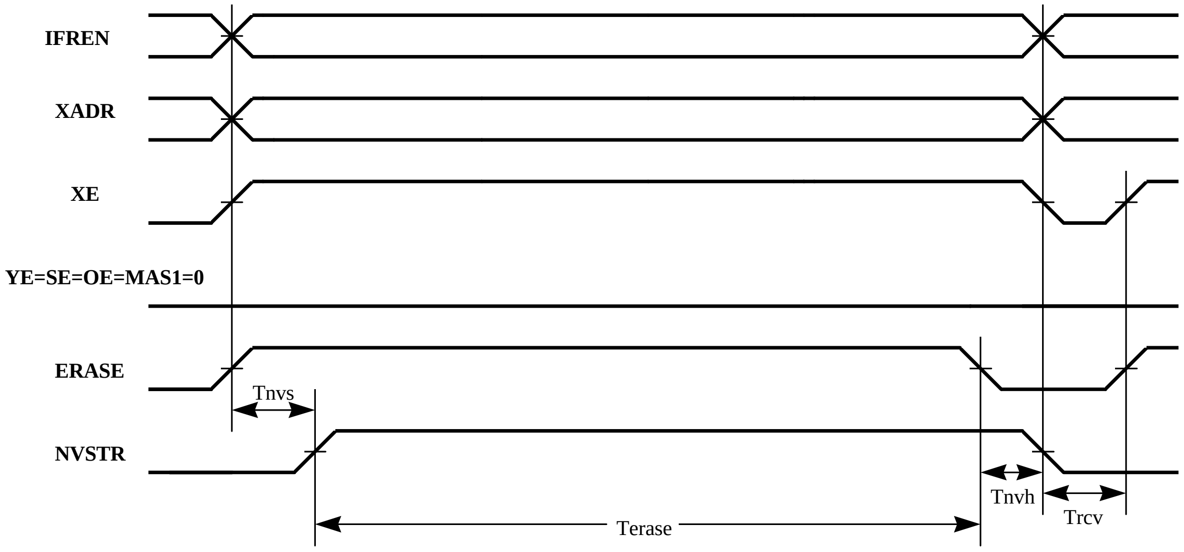
Read Cycle



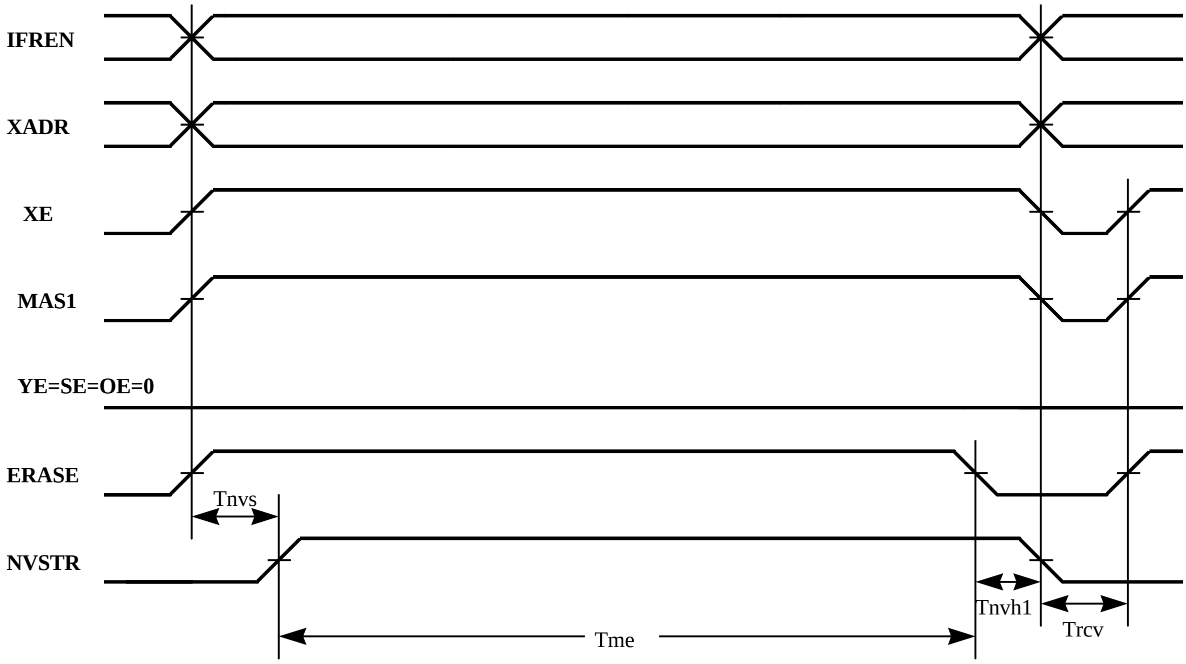
Program Cycle



Erase Cycle



Mass Erase Cycle



Update history:

Date	Rev	What	Who
07/26/2000	0.1	0.This new created spec. is modified from the original spec. of SFC0032_16B9 to meet the high endurance requirement. 1. The timing parameter of Tpgh is changed from 10ns to 20ns. 2. Size of IP: 724.388 x 1783.275 um^2	CHKuo